Physical Chemistry Cume
NMR Spectroscopy
April 18, 2008

(1) Draw the NMR spectrum of ^1H in the molecule, $^1H^2H$. The spin quantum number for ^1H is I=1/2 and that of ^2H is I=1.

(2) Predict the coupling constant, $^1J(HH)$, for the hydrogen molecule ($^1H_2$) from the following information. $^1J(^1H,^2H) = 43$ Hz. The magnetogyric ratios (γ) are 267.512×10^6 $radT^{-1}s^{-1}$ and 41.0648×10^6 $radT^{-1}s^{-1}$ for ^1H and ^2H, respectively. A one-bond coupling constant between nucleus X and nucleus Y is proportional to the product of the magnetiogryic ratios, $(\gamma)_x (\gamma)_y$, $^1J(X,Y) \propto (\gamma)_x (\gamma)_y$. Draw the experimentally obtained NMR spectrum of the $^1H_2$ molecule. Explain in detail this spectrum.

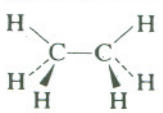
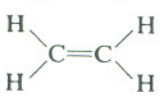
(3) The ^{13}C spectrum of $DCCl_3$ (D is ^2H) at 21.4 MHz provides $^1J(^{13}C,^2H) = 32.1$ Hz. From the table shown below, it can be seen that $^1J(^{13}C,^1H)$ is directly proportional to the fraction of s-character (f_s) in the carbon hybrid according to:

$$^1J(^{13}C,^1H) = (500 \text{ Hz})\,(f_s)$$

What is the fraction of s-character in the C-H bond in the $HCCl_3$ molecule ?

If the hybrid orbital is sp^n ,where $n=f_P/f_s$, what is the value of n for the C-H bond?

The Effect of Carbon Hybridization on $^1J_{CH}$[a]

Structure	Shape at carbon	C—C—H angle	Carbon hybrids	f_s	f_p	$^1J_{CH}$ (Hz)
	tetrahedral	109.5°	sp^3	0.25	0.75	125
	trigonal planar	120°	sp^2	0.33	0.67	156
H—C≡C—H	linear	180°	sp	0.50	0.50	249

[a] Note that $f_s + f_p = 1$.